

RES: A modular pipeline to recognise, encode and select relevant facts from clinical notes

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Conor Brown
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RES: A modular pipeline to recognise, encode and select relevant facts from clinical notes

Abstract—Clinical information extraction (IE) converts clinical narratives into structured facts. Large language models can do this with significantly less preprocessing than rules-based methods, but they introduce new classes of problems: hallucination, schema drift and overconfidence. We propose a hybrid system where the model recognises entities, interprets relations and provides evidence, and the rules normalise entities, verify evidence and apply policies to resolve conflicts. This modular system preserves the rich uncertainty captured in clinical narratives while producing auditable structured outputs that confirm to explicit rules. We use 2 public datasets of epilepsy letters to evaluate this system on two distinct tasks: (1) selecting the most important clinical fact among many and (2) building a broad inventory of clinical facts. We show that the combined system improved F1 by +0.05 vs. rules-only or LLM-only systems, while creating a rich evidence trail with an exact evidence score of ≥ 0.99 . This transparent, evidence-grounded system is a step towards making more trustworthy systems for clinical use.

Index Terms—information extraction, neuro-symbolic, evidence verification, natural language processing

I. INTRODUCTION

Clinical notes in electronic health records (EHRs) are among the most valuable sources of medical information available, recording diagnoses, medications, medical investigations and more in rich narrative form. This form is both a strength and a weakness. Patient demographics are routinely collected in structured forms for administrative purposes, and this structured form makes them easy to use in clinical research. Information about the patient’s diagnoses and treatment outcomes often exist solely in narrative form in hospital discharge summaries or letters from clinicians. This makes them useful to deeply understand an individual patient, but very difficult to use to build a broad understanding of a patient population.

Epilepsy is one of the most common neurological diseases, affecting over [x]m people globally. Clinicians use the type of epilepsy, type of seizures and frequency of seizures to determine the severity of the patient’s disease. [1] Clinical trials typically use seizure freedom as their target outcome. [2] The only place this information is routinely recorded is in clinic letters from an epilepsy specialist to the patient’s primary care doctor after the patient visits the clinic. To gather this information at scale, experts are required to complete manual chart review: review each letter, interpret the key details and translate them into a standard form. This process is costly, time-consuming, and error prone. [3] Making this information available in a structured form that you can query in a database to summarise, group and filter can make large-scale research efforts viable.

To make this type of structured data available, there are two broad approaches: collect the data in a structured form or process unstructured data into that form. It is simple to collect medication in a more structured form by using separate fields for medication, dose, unit and frequency, rather than writing ‘Topiramate 100 mg BD’. Applying that to seizure frequency is more difficult because they are not simple facts to record in a single, consistent form. A typical example is, *“focal impaired-awareness seizures have been occurring more often, 2 times per week, whereas previously they only happened once or twice a month.”* Imprecise language like this is known as hedging[5], and paradoxically, stripping it down to a single number would make it less precise and less useful. A system that enables clinicians to record these richer observations and then processes them into a standardised form is ideal.

Natural language processing (NLP) and the subdomain of information extraction (IE) offers a solution for this. Rules-based methods have successfully extracted information on medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency statements. ‘2 to 3 seizures a week’, ‘frequent bouts of seizures’ and ‘4 since her last visit’ are all valid descriptions of the same fact. Large language models (LLMs) have demonstrated significant improvements in semantic understanding than prior methods and have successfully extracted seizure frequency information, but they occasionally hallucinate facts. Combining the semantic understanding of an LLM with the transparency of a rules-based system could provide the most accurate, trustworthy results.

Effective IE can be broken down into three key stages: recognising relevant candidate facts, encoding those facts into a particular form, and selecting which facts to keep and which to filter out. A clinic letter can contain multiple descriptions of seizures, some are historical events while others are current, some are certain seizures while others may be “absence-like episodes”, some are describing the same event but in another context. Seizure frequencies can be expressed as a rate, a vague quantity and a duration. A transparent system would allow you to see not just the final answer, but which steps were taken to achieve it.

This paper evaluates the ideal combination of LLMs and rules across those three stages: recognition, encoding and selection. It was evaluated on two sets of annotated epilepsy clinic letters that reflect two common clinical IE tasks: producing a single label to determine a patient’s current status, and producing a broad inventory of clinical facts, past and present. The paper demonstrates that a modular system combining

LLMs and rules can extract information with comparable accuracy to past methods while preserving a richer evidence chain. The system is available as an open source package at [INSERT LINK](#). Figure 1 displays the web app illustrating some of that chain from evidence to final answer.

II. LITERATURE REVIEW

Epilepsy clinic letters contain facts about medication, diagnoses, investigations and seizure frequency, often covering multiple time periods and with many hedged [1] observations. Fonferko-Shadrach *et al.* treat the letter as an inventory of supported facts across prescriptions, diagnoses, investigations and frequency [1], [2]. Xie *et al.* [3], Holgate *et al.* [4], and Gan *et al.* [3] treat it as one current seizure-frequency state. The decisions about which facts to recognise, how to represent them and which facts to keep in the final representation are often bundled into one single task definition. This paper demonstrates that breaking them up into discrete stages with explicit subjective choices can make the system more transparent and flexible.

Wang *et al.* define clinical IE as the automatic extraction and encoding of clinical information from text [8]. Fu *et al.* separate mention detection from concept encoding: the quoted span is distinct from the coded concept [6]. Xie *et al.* extract a frequency sentence and then apply rules to encode it into a number [9]. In much of the remaining epilepsy literature encoding is bundled into the extraction phase, with the model extracting e.g. '4 per month' from a raw statement 'about once a week'. When rules are used to format the answers, they are often described as simple post-processing steps for scoring instead of distinct stages in the pipeline. Those paper shows that separating recognition and encoding as separate stages preserves more of the evidence while still being able to encode it into the required form for downstream use.

DECIDE WHAT TO SAY ABOUT IAA

A number of methods have shown performance at or above human level. Fonferko-Shadrach *et al.* reported the annotators found annotation difficult and seizure frequency was the most difficult. This was reflected in the inter annotator agreement (IAA) score of 0.47 (moderate agreement) even after multiple phases of development of the guidelines to meet the annotators' needs. Xie *et al.* reported an overall pairwise F1 score of 0.44 on a task focused extracting seizure freedom, frequency and most recent seizure statements and recommend double or triple annotation. This is a long-standing challenge in the field, as Grisham and Sundheim noted in 1996 'it proved remarkably difficult to formulate guidelines which were reasonably complete and consistent'. Despite levels of IAA of 0.5, the literature commonly reports that rules-based and LLM-based methods achieving F1 scores of 0.8 and above.

INSERT A REVIEW OF EVALUATION PERFORMANCE

Chiticariu *et al.* show that academic papers tend to focus exclusively on benchmark accuracy, while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy. Fu *et al.* show that this translates to the clinical setting too, and model interpretability is also a

critical factor in successful deployment in clinical settings. Clinicians are highly trained in using deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Fonferko-Shadrach *et al.* expose evidence spans as a consequence of using a rules-based system, and Gan *et al.* utilise evidence spans as a training signal. This paper presents a transparent evidence chain from exact text to final answer as part of the core output to improve interpretability for downstream users.

Most papers present results of a system trained or tuned on one local corpus. Decker *et al.* show that a highly tuned system on one set of epilepsy letters can perform significantly worse on a set from another epilepsy centre, with seizure-frequency F_1 dropping from 0.82 in-house to 0.40 at a second centre [12]. This is further compounded by the fact that because medical data is highly sensitive, most papers report results on private datasets that cannot be reproduced or evaluated on other systems. Fonferko-Shadrach *et al.* and Gan *et al.* are the two exceptions in this domain. This paper is the first to assess a method on two distinct tasks using two public datasets.

III. METHODS

A. Data

Two datasets were used in this paper. Both datasets are collections of epilepsy clinic letters which record a UK patient's consultation with an epilepsy specialist, often a neurologist or specialist nurse. [2], [3] The NHS requires that clinic letters are sent to the patient's GP after all outpatient visits such as these. [13] They are often the only record of a patient's diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. Both datasets are synthetic letters modelled on real clinic letters, and they were annotated by trained researchers and clinicians. An example clinic letter is shown in Figure 1.

The primary difference in the two datasets is breadth and depth of annotation. Fonferko-Shadrach *et al.* annotated a rich inventory of clinical facts, including the current anti-seizure medications used, the types of epilepsy and seizures the patient was diagnosed with, the investigations used for diagnosis and the frequency of seizures. Multiple medications, diagnoses, investigations and seizure frequency events can be recorded in a letter, and each clinical fact has multiple attributes describing it. Figure 1 illustrates an example of multiple seizure frequency events with multiple attributes linked to each. The task is to determine which clinical facts are relevant. This is referred to as the inventory dataset in this paper. Gan *et al.* annotates a single label per letter with the patient's current seizure frequency burden. The task is to determine which fact to choose among many. If Gan *et al.*'s annotation was applied to the letter in Figure 1, the label would be '1 per month'. This is referred to as the current frequency dataset.

The datasets also differ in how the letters were generated and labelled. Fonferko-Shadrach *et al.* produced 199 hand-authored letters by epilepsy specialists for the inventory

TABLE I: Dataset summary

	Dataset 1	Dataset 2
Annotated clinical categories	Medication, diagnosis, investigations and seizure frequency*	Seizure frequency
Letter generation and annotation method	Human-written letters and labels	AI-generated letters, human-written labels
Number of letters (dev/test)	199 (140/59)	1,200 (750/450)
Median words per letter (range)	217 (49-693)	400 (115-756)
Median number of annotated labels (range)	6 (0-14)	1 (1-1)

*The dataset includes 5 additional categories. This paper uses only these 4.

dataset. Gan *et al.* produced a much larger dataset of 15,099 letters using a combination of 10 templates and 500 short descriptions of seizure frequency produced by GPT-5 for the current frequency dataset. A sample of 1,200 letters from Gan *et al.* were used from current frequency, and all 199 letters from inventory. Table 1 summarises those details plus some statistics on the length of the letters and the number of labels per letter.

Minimal preprocessing was applied to the raw data. Dataset 1 stored the letters in raw .txt and annotations in .json files, while Dataset 2 stored both in a single .json file. Letters in both datasets are read in UTF-8 format and minimally cleaned to correctly parse quotes, new lines, tabs, the $\leq$ sign and nulls. The raw letters were ingested in the pipeline in full; no tokenisation, section-splitting or truncation was applied.

Datasets were split up into stratified samples for separate development and test sets. Inventory was divided into 140 development/59 test sets. They were stratified by the existence of a seizure frequency mention, so c. 72% of letters in both splits contain a seizure frequency fact. Current frequency was divided into 750 development letters and 450 test letters. They were stratified by the kind of seizure frequency fact so both sets consist of 62% with a frequency, 15% seizure free, 13% unknown and 10% with no seizure frequency reference. Seizure frequency was used to determine the stratification because it is the most important clinical outcome for epilepsy patients and the only shared label between both datasets.

B. Architecture

To trace the journey from raw evidence to final answer, the task was divided into three key stages: recognition, encoding, selection. First, the pipeline identifies potential candidates that fit the description in the raw text. Next, the pipeline takes the raw text from those candidates and encodes them into the required form. Finally, the pipeline determines which of those encoded facts meet the criteria for the final output. The specific configuration of the each stage is determined by the nature of the task: which labels to extract from the text, what schema to encode them into, and how many labels to output at the final stage. The same modular architecture effectively describes multi-label tasks such as inventory and single label

TABLE II: Pipeline configurations

Configuration	Recognise	Encode	Select
Rules only	Rules	Rules	Rules
LLM + Rules	LLM	LLM + Rules	Rules
LLM only	LLM	LLM	LLM

tasks such as current frequency. Figure 2 illustrates the general architecture of the pipeline.

There are two key benefits of breaking the task into three discrete stages: configurability and transparency. Past research has shown that entirely rules-based methods can successfully populate the inventory, while entirely LLM-based methods can successfully extract the current frequency. While some methods have used a combination of rules and LLMs to produce the final answer, none have empirically assessed the ideal combination of the two. This three-stage pipeline enables evaluation of the best combination overall, and specifically which component performs best at which stage. Table 2 summarises the 3 main configurations of the pipeline evaluated in this paper.

C. Study Design

The primary objective of this research is to identify which component performs best at which stage. Table 2 shows the 3 pipeline configurations used in the evaluation: rules at every stage, the LLM at every stage or a combination of LLM and rules. For each configuration we evaluated the pipeline at each stage: recognise, encode and select. This will produce a 3x3 matrix of results that answer two key questions:

- 1) Among the three pipeline configurations, which performs best: rules, LLM, or a combination of both?
- 2) How much incremental value is there from the additional encode and select stages?

Evaluating the pipeline on both datasets provides insight into whether this method generalises to both single and multi-label prediction tasks with distinctly different challenges. Past research has shown that individual methods which perform well on extracting a broad clinical inventory perform worst on extracting seizure frequency, and methods that extract seizure frequency effectively on one corpus generalise poorly to another corpus with a different structure and writing style.

For the inventory task, the challenging part is finding the right balance between precision and recall. On average there are 6 labels per letter to extract, but some letters have 0 and some have 14. In this scenario it would be easy to tune a pipeline that either over-extracts when there is an absence of real evidence(false positives) or under-extracts when there is an abundance of evidence (false negatives). The benefit of breaking the task into three stages is that you can then tune each stage accordingly. The candidate recognition phase can be tuned to maximise recall at the cost of some precision, recognising a high number of true positives with a reasonable number of false positives. The selection phase can then be tuned for precision to filter out the false positives. We can then evaluate whether LLMs or rules perform best at each task.

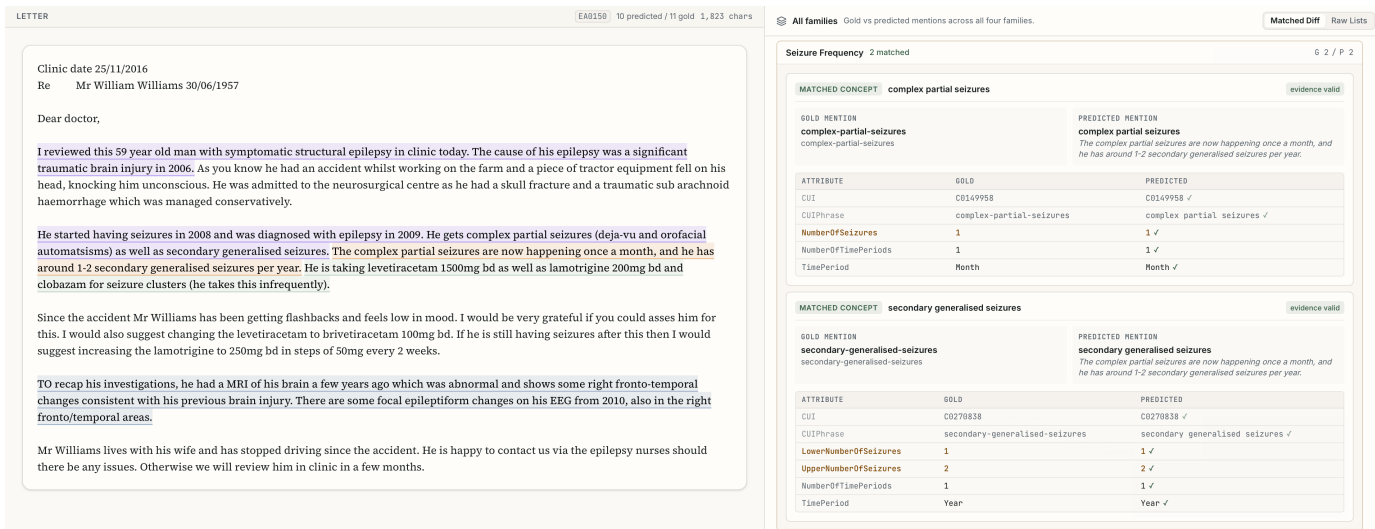


Fig. 1: The left pane shows a clinic letter from Fonferko-Shadrach *et al.* annotated for medications, diagnoses, investigations and seizure frequencies. The right pane illustrates how multiple seizure events are represented in the schema.

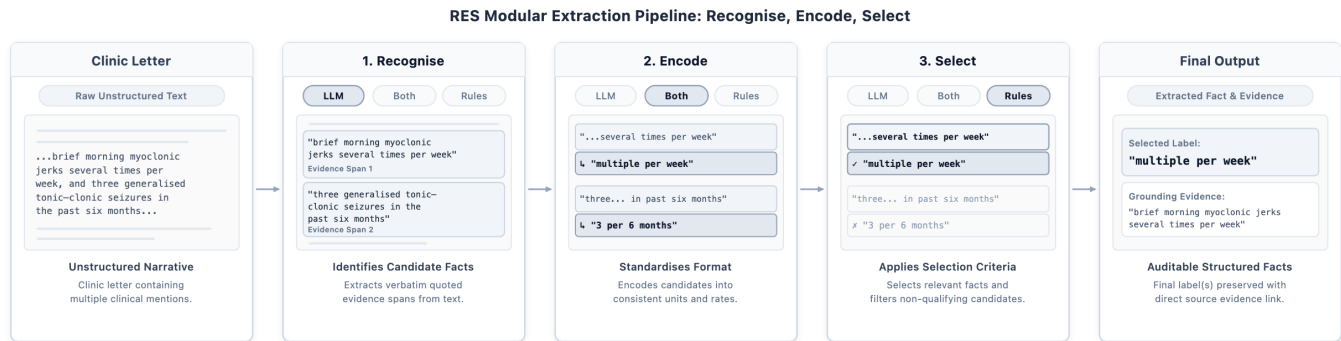


Fig. 2: A clinic letter passes through three stages in the pipeline - recognise, encode and select - to generate structured labels with the exact text used from the letter to produce the label. Each of the three stages can be configured to use an LLM, rules or both. The example shown is the task of extracting a single label representing the patient's current seizure frequency burden.

For the current frequency dataset, the challenging part is determining which one fact among many is the correct answer. Sometimes that requires grouping together multiple related facts into a single answer, e.g. '1 in June, 0 in July, 2 in August' becomes '3 in 3 months'. Sometimes it requires determining which fact takes precedence when there are multiple competing facts with varying degrees of specificity and certainty e.g. 'She may remain seizure-free for up to 4 month, but then will experience clusters of 5 seizures in a single day'. Different annotation guidelines would record ambiguous facts like these differently. Distinct stages in the pipeline make those choices configurable and explicit.

The primary hypothesis is that a combination of an LLM plus rules will perform better than either on their own, because they have complementary strengths and those strengths are particularly well suited to distinct stages in the pipeline. The secondary hypothesis is that multiple discrete stages will improve performance for the LLM, whether the subsequent

stages are rules or additional LLM calls. This builds on the idea that task decomposition is an effective form of context engineering, and is particularly useful in scenarios where the prompt is long and contains many layers of instructions.

Splitting the pipeline into stages is therefore designed to make it easier to evaluate and easier to optimise, producing better results overall.

D. Pipeline Optimisation

Over 50 evaluations were conducted on the development set to optimise the two key elements of the pipeline: the LLM prompt instructions and the rules. The focus of the optimisation process was to optimise accuracy, precision, recall or the F1 score, depending on the task and stage. In-depth qualitative error analysis was conducted to diagnose the cause of poor performance in particular areas and identify common error modes. Hypotheses were developed and tested to tune the prompt and rules iteratively.

Multiple configurations of the overall architecture were explored in the early stages, particularly for the LLM component. Three distinct architectures were:

- 1) Task specialisation: breaking up the inventory task into individual prompts for each clinical family.
- 2) Document splitting: splitting up the document into discrete sections and providing targeted instructions for each section.
- 3) Generator-verifier: using the first LLM to provide the answers and an LLM-as-a-judge to verify them.

All configurations were based on the assumption that the LLM could provide the best answer in a single prompt, as this was the common design in the literature. Through iterative error analysis on both the LLM and the rules, it became apparent that the complexity of the task was more often a function of competing rules and guidelines on the same clinical family, rather than the breadth of rules across families. Often improving a rule or a prompt instruction to address one class of error would harm another area because annotation policies are complex and can appear contradictory in some instances. This complexity is a challenge for human annotators also, as described above in the literature review.

Decomposing the task into three stages allowed for cleaner separation of the policies on which types of facts to extract and which criteria to apply to exclude specific variants of them. Decomposing the task into three stages also made it possible to evaluate the performance at each individual and clear identify the bottlenecks in the system. It is this combination of a cleaner task decomposition for the pipeline and a clearer view into the pipeline that enabled steady, incremental gains to materialise from the optimisation process.

E. Evaluation

The pipeline was evaluated using a single headline metric for each dataset: the micro-F1 score for inventory and accuracy for current frequency. The reason the two datasets use two different headline metrics is because current frequency only contains a single label for each letter, while inventory contains an average of 6 labels per letter. Simple accuracy is sufficient for a single label, but multi-label predictions require a more sophisticated metric to evaluate performance.

Accuracy measures how often the correct answer was selected out of all answers provided. In the current frequency task, there is only one answer per letter so accuracy is a complete measurement of how often the pipeline selected the correct answer. Before computing the accuracy, Gan *et al.* categorised the monthly frequency rates into two groups: a fine-grained category called 'purist' and a coarse category called 'pragmatic'. For example, the label '1 per month' was assigned to the purist group 'monthly, not weekly' and the pragmatic group 'frequent'. This paper uses purist accuracy as the primary metric for current frequency because distinguishing between daily and monthly seizures is a clinically important detail.

The micro-F1 score is used for inventory as it measures how well the pipeline can accurately extract facts from all

four clinical categories. The F1 score measures the harmonic mean of precision and recall. Recall measures how many of the correct answers can be predicted, while precision measures how often a prediction is correct. Micro-F1 takes the average F1 score across each of the categories, in this case medication, diagnosis, investigations and seizure frequency. The macro-F1 score would assign more weight to the diagnosis category because it contains more labels per letter. The micro-F1 score ensures each category is given equal importance.

All metrics presented in the paper use the held-out test sets. This measures how well the pipeline generalises to examples it has not seen before. The development sets were used explicitly to optimise the rules and prompts through iterative error analysis and performance can be inflated by over-fitting to specific patterns in those letters. The held-out test set is used to reduce that risk and evaluate how well the pipeline can identify meaningful patterns. Letters from the development set are used in this paper to show how a prediction moves through each of the stages in the pipeline and the evidence chain it is connected to, or to illustrate a particular type of error.

IV. RESULTS

A. Development Environment

B. Methods

For the inventory task the three-stage lens made it clear that the major limiting factor was low recall at the recognition phase. Precision was consistently high but the prompt instructions were causing too many valid candidates to be excluded in a single extraction call. Using more permissive guidelines in the recognition phase significantly improved recall at the cost of some precision, but the select phase was able to recover that precision.

C. Models

D. Structured Evidence

E. Ablations

F. Error Analysis

V. DISCUSSION

VI. CONCLUSION

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2].

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